1 Systems of Linear Equations (SLE) with Unknown Moduli

A recurring vulnerability in the examined schemes is the susceptibility to attacks via Systems of Linear Equations (SLE). However, a fundamental theoretical question arises: Is it mathematically feasible to solve an SLE when the underlying composite modulus M is unknown to the attacker?

When attempting to solve an SLE modulo a composite number M (where $M = P \cdot Q$ via the Chinese Remainder Theorem), one is essentially solving two independent linear systems simultaneously:

$$P \cdot x = i \tag{1}$$

$$Q \cdot y = j \tag{2}$$

If an adversary only knows one of the two variables, solving the remaining system seems inherently restricted. Assuming an attacker attempts to solve $M \cdot x = i$, they are effectively attempting to satisfy:

$$P \cdot x = i \tag{3}$$

$$Q \cdot x = i \tag{4}$$

If the sub-system modulo Q has no valid solution, the overarching system modulo M is consequently unsolvable, as a valid solution must satisfy both systems simultaneously.

1.1 The Prime Factorization (PFZ) Leak

The primary condition that renders such a modular SLE unsolvable is the necessity to divide by zero (or encountering a zero-divisor). However, if the scheme leaks a zero-divisor to the adversary, it inadvertently allows the extraction of the prime factorization (PFZ) of M (e.g., via $\gcd(x, M)$).

Even if evaluated in a complex algebraic space where leaking a null-vector might seem benign (assuming the null-vector itself is not used for decryption), exposing any zero-equivalent structure mathematically guarantees the compromise of the prime factors.

2 Checksums and Length Encoding

To circumvent the SLE vulnerability, a specific encoding strategy was proposed: Utilizing a length encoding for the message m, accompanied by a homogeneous checksum vector $[r, r, \ldots, r]$.

By applying the Chinese Remainder Theorem with orthogonal idempotents p_1q_0 and p_0q_1, the ciphertext c is constructed as:

$$c := [m_0 g^0, m_1 g^1, \ldots, m_n g^n] \cdot p_1 q_0 + [r g^0, r g^1, \ldots, r g^n] \cdot p_0 q_1 \tag{5}$$

The rationale is that the checksum component does not possess full matrix rank. Under cyclic convolution (homomorphic multiplication), the checksum part remains linearly dependent upon itself:

$$[r g^0, \ldots, r g^n] * [r g^0, \ldots, r g^n] = [r^2 g^0, \ldots, r^2 g^n] = k \cdot r \cdot [r g^0, \ldots, r g^n] \tag{6}$$

Because the checksum component is strictly linearly dependent, the SLE cannot be fully resolved for this part.

2.1 Failure of the Checksum Strategy

Despite the mathematical soundness of the rank reduction, this approach remains insecure. To prevent the SLE from being solved, the adversary must encounter a complex number without a multiplicative inverse. This occurs exclusively when the norm evaluates to zero, or if the element shares a common factor with the composite group order. In either scenario, the structural failure directly leaks the prime factorization (PFZ) of the modulus.

3 Conclusion on Exact Linearity

Based on these iterative proofs and counterexamples, a definitive conclusion can be drawn:

Theorem (Conjecture): The combination of exact linear transformations and modular arithmetic cannot yield a secure Fully Homomorphic Encryption (FHE) scheme. Any mechanism that successfully prevents Gaussian elimination by introducing non-invertible elements inherently exposes the prime factorization of the underlying modulus.

4 Future Work: Non-Linear Constructions

To achieve secure homomorphic operations, the reliance on linearity must be abandoned. A potential non-linear solution could be based on orthogonal idempotent pairs (z, z'). The concept involves utilizing (z, z') to encode two independent real numbers, subsequently obfuscated through a Discrete Fourier Transform (DFT).